Quarkyonic Matter Diego's Notes

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Notes from Dr Vovchenko QM and group papers

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Contents

Particle Number Fluctuations for van der Waals Equation of State

1.1 From Ideal gas to the van der Waals EOS

We begin with the ideal gas law,

$$PV = nRT. \tag{1}$$

This equation assumes:

- the particles have negligible volume,
- there are no intermolecular forces between particles.

These assumptions work well at low density, but they fail for real gases.

Therefore, to construct a more realistic equation of state, we introduce two corrections:

i. a correction for the **finite size** of the particles,

ii. a correction for the **attractive interactions** between particles.

i) Volume correction: finite molecular size

In the ideal gas model, particles are treated as point-like, so they can move in the full volume V.

For a real gas, however, molecules occupy space, so the volume actually available for motion is smaller than V.

If b is the excluded volume per mole, then for n moles the unavailable volume is approximately nb. Hence, the effective free volume becomes

$$V_{\text{free}} = V - nb. \tag{2}$$

Replacing V by $V - nb$ in the ideal gas expression gives

$$P \approx \frac{nRT}{V - nb}. \tag{3}$$

This is the first correction.

ii) Pressure correction: attractive interactions

In a real gas, molecules attract each other.

Because of this, a molecule approaching the wall is pulled slightly inward by neighboring molecules. As a consequence, the force exerted on the wall is reduced, and the observed pressure is smaller than it would be in the absence of attractions.

Thus, the pressure appearing in the ideal-like expression must be larger than the measured pressure P. We write

$$P_{\text{ideal-like}} = P + \Delta P. \tag{4}$$

Now we estimate ΔP.

The strength of the attractive correction should increase with the number of interacting pairs. Since the number density is

$$\rho = \frac{n}{V}, \tag{5}$$

the number of pair interactions scales as ρ^2. Therefore,

$$\Delta P \propto \left(\frac{n}{V}\right)^2. \tag{6}$$

Introducing a proportionality constant a, which measures the strength of the attractive interactions, we obtain

$$\Delta P = a\left(\frac{n}{V}\right)^2 = a\frac{n^2}{V^2}. \tag{7}$$

Hence,

$$P_{\text{ideal-like}} = P + a\frac{n^2}{V^2}. \tag{8}$$

Combining both corrections

Now we replace:

- ★ $V \to V - nb$,
- ★ $P \to P + a\frac{n^2}{V^2}$.

Then the ideal gas law becomes

$$\boxed{\left(P + a\frac{n^2}{V^2}\right)(V - nb) = nRT}, \tag{9}$$

this is the **van der Waals equation of state**, or:

$$\boxed{p(V,T,N) = \frac{NT}{V - bN} - a\frac{N^2}{V^2}}. \tag{10}$$

1.2 From CE to GCE

In the GCE the macroscopic states of a thermal system are defined in terms of the grand canonical potential Ω which is function of three physical variables (volume V, temperature T abd chemical potential μ). It is conected to the system pressure as:

$$\Omega(V,T,\mu) = -p(T,\mu)\,V. \tag{11}$$

Therefore, pressure asa function of its natural variables, gives a complete description of the eos, one calculates the entropy S, (conserved) number density n, and energy density ϵ as:

$$S(T,\mu) = \left(\frac{\partial p}{\partial T}\right)_\mu, \qquad n(T,\mu) = \left(\frac{\partial p}{\partial \mu}\right)_T, \qquad \epsilon(T,\mu) = TS + \mu n - p. \tag{12}$$

The thermodynamical potential in the CE is the free energy $F(V,T,N)$. The function F, depends on its natural variables (T,V,N) and it gives a complete thermodynamical description of the system properties. Entropy, pressure, chemical potetnial and the total energy can be calculated in the CE as:

$$S(V,T,N) = -\left(\frac{\partial F}{\partial T}\right)_{V,N}, \qquad p(V,T,N) = -\left(\frac{\partial F}{\partial V}\right)_{T,N}, \qquad \mu(V,T,N) = \left(\frac{\partial F}{\partial N}\right)_{T,V} \tag{13}$$

and

$$F + TS \tag{14}$$

If one knows pressure in the CE it is not yet possible to find entropy, chemical potential and energy in unique way. For a complete thermodynamical description in CE one needs to reconstruct free energy, thus, the following should be solved:

$$\left(\frac{\partial F}{\partial V} = -\, p(V, T, N) \right). \tag{15}$$

The solution of later eq can derived as:

$$F(V, T, N) = F(V_0, T, N) - \int_{V_0}^{V} p(V', T, N)\, dV \tag{16}$$

To find $F(V, T, N)$ one needs to know not only pressure but, additionally, the free energy $F(V_0, T, N)$ as the function of T and N at some fixed value $V = V_0$. We assume that for fixed T the role of particle interactions becomes neglible at very small density $N/V_0 \to 0$. For relativistic ideal Boltzmann gas the free energy reads

$$F_{\mathrm{id}}(V, T, N) = -NT \left[1 + \ln\!\left(\frac{V\, \phi(T; d, m)}{N} \right) \right] \tag{17}$$

with

$$\boxed{\phi(T; d, m) = \frac{d}{2\pi^2} \int_0^\infty k^2\, dk\, \exp\left[-\frac{\sqrt{k^2 + m^2}}{T} \right] = \frac{d\, m^2 T}{2\pi^2} K_2\!\left(\frac{m}{T} \right)}. \tag{18}$$

Full derivation from later eq:
We begin from the canonical ensemble relation

$$F_{\mathrm{id}}(V, T, N) = -T \ln Z(V, T, N), \tag{19}$$

where $Z(V, T, N)$ is the canonical partition function.
For a gas of N identical, non-interacting classical particles,

$$Z(V, T, N) = \frac{1}{N!} \left[Z_1(V, T) \right]^N, \tag{20}$$

where $Z_1(V, T)$ is the one-particle partition function.
For a relativistic particle with energy

$$\varepsilon(k) = \sqrt{k^2 + m^2}, \tag{21}$$

the one-particle partition function is

$$Z_1(V, T) = d \int \frac{d^3x\, d^3k}{(2\pi)^3} \exp\left[-\frac{\sqrt{k^2 + m^2}}{T} \right]. \tag{22}$$

Here, d is the degeneracy factor. Since the integrand does not depend on position,

$$\int d^3x = V, \tag{23}$$

thus

$$Z_1(V, T) = V\, d \int \frac{d^3k}{(2\pi)^3} \exp\left[-\frac{\sqrt{k^2 + m^2}}{T} \right]. \tag{24}$$

Now define

$$\phi(T;d,m) = d\int \frac{d^3k}{(2\pi)^3}\,\exp\left[-\frac{\sqrt{k^2+m^2}}{T}\right]. \tag{25}$$

Then

$$Z_1(V,T) = V\,\phi(T;d,m), \tag{26}$$

and therefore

$$Z(V,T,N) = \frac{[V\,\phi(T;d,m)]^N}{N!}. \tag{27}$$

Using

$$F_{\mathrm{id}}(V,T,N) = -T\ln Z(V,T,N), \tag{28}$$

we get

$$\begin{aligned} F_{\mathrm{id}}(V,T,N) &= -T\ln\left(\frac{[V\phi(T;d,m)]^N}{N!}\right) && (29)\\ &= -T\left[N\ln\big(V\phi(T;d,m)\big) - \ln N!\right]. && (30)\end{aligned}$$

For large N, we use Stirling's approximation,

$$\ln N! \approx N\ln N - N. \tag{31}$$

Substituting this into the free energy expression,

$$\begin{aligned} F_{\mathrm{id}}(V,T,N) &= -T\left[N\ln\big(V\phi(T;d,m)\big) - N\ln N + N\right] && (32)\\ &= -NT\left[\ln\big(V\phi(T;d,m)\big) - \ln N + 1\right]. && (33)\end{aligned}$$

Using

$$\ln A - \ln B = \ln\left(\frac{A}{B}\right), \tag{34}$$

we obtain

$$\boxed{F_{\mathrm{id}}(V,T,N) = -NT\left[1+\ln\left(\frac{V\,\phi(T;d,m)}{N}\right)\right]}. \tag{35}$$

We now evaluate

$$\phi(T;d,m) = d\int \frac{d^3k}{(2\pi)^3}\,\exp\left[-\frac{\sqrt{k^2+m^2}}{T}\right]. \tag{36}$$

Since the integrand depends only on the magnitude $k = |\mathbf{k}|$, we use spherical coordinates in momentum space:

$$d^3k = 4\pi k^2\,dk. \tag{37}$$

Thus,

$$\begin{aligned} \phi(T;d,m) &= d\frac{4\pi}{(2\pi)^3}\int_0^\infty k^2\,dk\,\exp\left[-\frac{\sqrt{k^2+m^2}}{T}\right] && (38)\\ &= \frac{d}{2\pi^2}\int_0^\infty k^2\,dk\,\exp\left[-\frac{\sqrt{k^2+m^2}}{T}\right]. && (39)\end{aligned}$$

Hence,

$$\boxed{\phi(T;d,m) = \frac{d}{2\pi^2}\int_0^\infty k^2\,dk\,\exp\left[-\frac{\sqrt{k^2+m^2}}{T}\right]}. \tag{40}$$

Consider the integral

$$I = \int_0^\infty k^2\,dk\,\exp\left[-\frac{\sqrt{k^2+m^2}}{T}\right]. \tag{41}$$

Introduce the change of variables

$$k = m\sinh\theta, \qquad \sqrt{k^2+m^2} = m\cosh\theta, \qquad dk = m\cosh\theta\,d\theta. \tag{42}$$

Then

$$k^2\,dk = m^2\sinh^2\theta\,(m\cosh\theta\,d\theta) = m^3\sinh^2\theta\cosh\theta\,d\theta. \tag{43}$$

Therefore,

$$I = m^3\int_0^\infty \sinh^2\theta\,\cosh\theta\,e^{-(m/T)\cosh\theta}\,d\theta. \tag{44}$$

Now use

$$\sinh^2\theta = \cosh^2\theta - 1, \tag{45}$$

so that

$$\sinh^2\theta\,\cosh\theta = \cosh^3\theta - \cosh\theta. \tag{46}$$

Thus,

$$I = m^3\int_0^\infty \big(\cosh^3\theta - \cosh\theta\big)\,e^{-(m/T)\cosh\theta}\,d\theta. \tag{47}$$

Using the identity

$$\cosh 3\theta = 4\cosh^3\theta - 3\cosh\theta, \tag{48}$$

we solve for $\cosh^3\theta$:

$$\cosh^3\theta = \frac{1}{4}\left(\cosh 3\theta + 3\cosh\theta\right). \tag{49}$$

Hence,

$$\cosh^3\theta - \cosh\theta = \frac{1}{4}\left(\cosh 3\theta + 3\cosh\theta\right) - \cosh\theta \tag{50}$$

$$= \frac{1}{4}\left(\cosh 3\theta - \cosh\theta\right). \tag{51}$$

Therefore,

$$I = \frac{m^3}{4}\int_0^\infty \left(\cosh 3\theta - \cosh\theta\right)e^{-(m/T)\cosh\theta}\,d\theta. \tag{52}$$

Now use the integral representation of the modified Bessel function of the second kind,

$$K_\nu(z) = \int_0^\infty e^{-z\cosh t}\cosh(\nu t)\,dt. \tag{53}$$

Then

$$I = \frac{m^3}{4}\left[K_3\Big(\frac{m}{T}\Big) - K_1\Big(\frac{m}{T}\Big)\right]. \tag{54}$$

Next, apply the recurrence relation

$$K_{\nu+1}(z) - K_{\nu-1}(z) = \frac{2\nu}{z}K_\nu(z). \tag{55}$$

Setting $\nu = 2$, we obtain

$$K_3(z) - K_1(z) = \frac{4}{z}K_2(z). \tag{56}$$

With $z = m/T$, this gives

$$I = \frac{m^3}{4} \cdot \frac{4T}{m} K_2\Big(\frac{m}{T}\Big) = m^2 T\, K_2\Big(\frac{m}{T}\Big). \tag{57}$$

Thus,

$$\boxed{\int_0^\infty k^2\, dk\, \exp\left[-\frac{\sqrt{k^2+m^2}}{T}\right] = m^2 T\, K_2\Big(\frac{m}{T}\Big)}. \tag{58}$$

Substituting into $\phi(T; d, m)$,

$$\phi(T; d, m) = \frac{d}{2\pi^2}\, m^2 T\, K_2\Big(\frac{m}{T}\Big). \tag{59}$$

Hence,

$$\boxed{\phi(T; d, m) = \frac{d\, m^2 T}{2\pi^2} K_2\Big(\frac{m}{T}\Big)}. \tag{60}$$

Combining everything,

$$\boxed{F_{\rm id}(V, T, N) = -NT\left[1 + \ln\Big(\frac{V\,\phi(T; d, m)}{N}\Big)\right]} \tag{61}$$

with

$$\boxed{\phi(T; d, m) = \frac{d}{2\pi^2}\int_0^\infty k^2\, dk\, \exp\left[-\frac{\sqrt{k^2+m^2}}{T}\right] = \frac{d\, m^2 T}{2\pi^2} K_2\Big(\frac{m}{T}\Big)}. \tag{62}$$

In eqs 17 and 18, d is the degeneracy factor, m is the particle mass and quantity ϕ has a physical meaning of the GCE particle number density at zero chemical potential.

Substituting in eq 16

$$F(V,T,N) = -NT\left[1 + \ln\frac{V_0\phi(T:d,m)}{N}\right] + \int_V^{V_0} \frac{NT}{V' - bN} \tag{63}$$

$$F(V,T,N) = -NT\left[1 + \ln\frac{V_0\phi(T:d,m)}{N}\right] + NT\int_V^{V_0} \frac{1}{V' - bN} \tag{64}$$

let be $u = V' - bN$, the limits become

$$V_0 - bN = u_0$$

and

$$V - bN = u$$

$$F(V,T,N) = -NT\left[1 + \ln\frac{V_0\phi(T:d,m)}{N}\right] + NT\int_u^{u_0} \frac{1}{u} \tag{65}$$

$$F(V,T,N) = -NT\left[1 + \ln\frac{V_0\phi(T:d,m)}{N}\right] + NT\ln u|_{V-bN}^{V_0-bN} \tag{66}$$

$$F(V,T,N) = -NT\left[1 + \ln\frac{V_0\phi(T:d,m)}{N}\right] + NT\ln\frac{V_0 - bN}{V - bN} = F_{id}(V - bN, T, N) \tag{67}$$

For the free energy (later eq), the chemical potential can be calculated as:

$$F(V,T,N) = -NT\left[1 + \ln\Big(\frac{V_0\,\phi(T;d,m)}{N}\Big)\right] + NT\ln\Big(\frac{V_0 - bN}{V - bN}\Big) = F_{\rm id}(V - bN, T, N) \tag{68}$$

For the free energy above, the chemical potential is

$$\mu = \left(\frac{\partial F}{\partial N}\right)_{V,T}.$$

Starting from

$$F = -NT\left[1 + \ln\left(\frac{V_0\phi(T;d,m)}{N}\right)\right] + NT\ln\left(\frac{V_0 - bN}{V - bN}\right),$$

we differentiate term by term:

$$\mu = \frac{\partial}{\partial N}\left[-NT\left(1 + \ln\left(\frac{V_0\phi(T;d,m)}{N}\right)\right)\right] + \frac{\partial}{\partial N}\left[NT\ln\left(\frac{V_0 - bN}{V - bN}\right)\right]. \tag{69}$$

For the first term,

$$\begin{aligned}
\frac{\partial}{\partial N}\left[-NT\left(1 + \ln\left(\frac{V_0\phi(T;d,m)}{N}\right)\right)\right] &= -T\left(1 + \ln\left(\frac{V_0\phi(T;d,m)}{N}\right)\right) - NT\frac{\partial}{\partial N}\ln\left(\frac{V_0\phi(T;d,m)}{N}\right) && (70)\\
&= -T\left(1 + \ln\left(\frac{V_0\phi(T;d,m)}{N}\right)\right) - NT\left(-\frac{1}{N}\right) && (71)\\
&= -T\ln\left(\frac{V_0\phi(T;d,m)}{N}\right). && (72)
\end{aligned}$$

For the second term,

$$\begin{aligned}
\frac{\partial}{\partial N}\left[NT\ln\left(\frac{V_0 - bN}{V - bN}\right)\right] &= T\ln\left(\frac{V_0 - bN}{V - bN}\right) + NT\frac{\partial}{\partial N}\ln\left(\frac{V_0 - bN}{V - bN}\right) && (73)\\
&= T\ln\left(\frac{V_0 - bN}{V - bN}\right) + NT\left[\frac{-b}{V_0 - bN} - \left(\frac{-b}{V - bN}\right)\right] && (74)\\
&= T\ln\left(\frac{V_0 - bN}{V - bN}\right) + bNT\left(\frac{1}{V - bN} - \frac{1}{V_0 - bN}\right). && (75)
\end{aligned}$$

Therefore,

$$\mu = -T\ln\left(\frac{V_0\phi(T;d,m)}{N}\right) + T\ln\left(\frac{V_0 - bN}{V - bN}\right) + bNT\left(\frac{1}{V - bN} - \frac{1}{V_0 - bN}\right). \tag{76}$$

where we have used that $(V_0 - bN)/V_0 \to 1$ at $V_0 \to \infty$

$$\mu = -T\ln\left(\frac{(V - bN)\phi(T;d,m)}{N}\right) + \frac{bNT}{V - bN}. \tag{77}$$

For particle number density $n = N/V$ one finds that for the ideal gas, set $b = 0$ in Eq. (77). Then

$$\mu = -T\ln\left(\frac{V\,\phi(T;d,m)}{N}\right). \tag{78}$$

Hence

$$\frac{\mu}{T} = -\ln\left(\frac{V\,\phi(T;d,m)}{N}\right), \tag{79}$$

and exponentiating both sides gives

$$\exp\left(\frac{\mu}{T}\right) = \frac{N}{V\,\phi(T;d,m)}. \tag{80}$$

Therefore,

$$\frac{N}{V} = \exp\left(\frac{\mu}{T}\right)\phi(T;d,m) \equiv n_{\mathrm{id}}(T,\mu). \tag{81}$$

On the other hand at $b > 0$ easily we have:

2 Van der Waals Equation of State with Fermi Statistics for Nuclear Matter

3 Quantum van der Waals theory meets quarkyonic matter

Mixture of Nucleons and Quarks with Pauli Exclusion Principle

The densities of baryon number, electric charge, and energy contain contributions from quarks and baryons and read

$$\rho_B = n_p^{\rm id}(k_F^p) + n_n^{\rm id}(k_F^n) + n_u^{\rm id}(q_F^u) + n_d^{\rm id}(q_F^d),$$
$$\rho_Q = n_p^{\rm id}(k_F^p) + 2\,n_u^{\rm id}(q_F^u) - n_d^{\rm id}(q_F^d),$$
$$\epsilon = \varepsilon_p^{\rm id}(k_F^p) + \varepsilon_n^{\rm id}(k_F^n) + \varepsilon_u^{\rm id}(q_F^u) + \varepsilon_d^{\rm id}(q_F^d).$$

After formualte the pauli principle we have that:

$$n_u = \frac{1+y}{2-y}\,\frac{g}{2\pi^2}\int_0^{h_{b,u}/N_c} q\sqrt{q^2+\Lambda^2}\,dq,$$

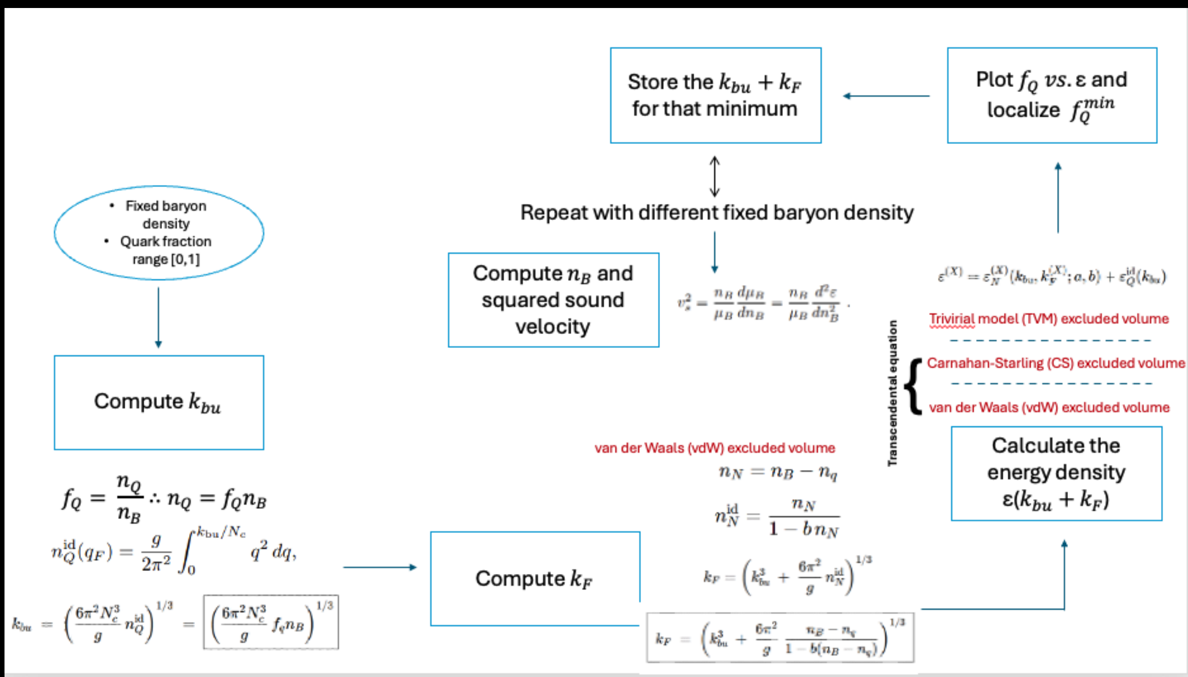
$$n_d = \frac{g}{2\pi^2}\int_0^{h_{b,u}/N_c} q\sqrt{q^2+\Lambda^2}\,dq.$$

The ideal gas functions are given as

$$n_{p,n}^{\rm id}(k_F^{p,n}) = \frac{g}{2\pi^2}\int_0^{k_F^{p,n}} k^2 \rho_{p,n}(k)\,dk,$$
$$\varepsilon_{p,n}^{\rm id}(k_F^{p,n}) = \frac{g}{2\pi^2}\int_0^{k_F^{p,n}} k^2 \rho_{p,n}(k)\,\varepsilon_{p,n}(k)\,dk,$$
$$n_{u,d}^{\rm id}(q_F^{u,d}) = \frac{g}{2\pi^2}\int_0^{q_F^{u,d}} q^2 \rho_{u,d}(q)\,dq,$$
$$\varepsilon_{u,d}^{\rm id}(q_F^{u,d}) = N_c\,\frac{g}{2\pi^2}\int_0^{q_F^{u,d}} q^2 \rho_{u,d}(q)\,\varepsilon_{u,d}(q)\,dq.$$

Meanwhile for the ideal gas we got the fermi momentum expressed as:

$$n_{p,n}^{\rm id} = \frac{g}{2\pi^2}\int_{k_{bu}}^{k_F^{p,n}} k^2\,dk = \frac{g}{6\pi^2}\left[(k_F^{p,n})^3 - k_{bu}^3\right].$$



4 Quantum van der Waals quarkyonic matter at nonzero isospin asymmetry

5 Correlations between nuclear incompressibility, liquid-gas critical point, and quarkyonic transition

5.1 Ground state of nuclear matter

The model parameters are constrained by nuclear ground state properties. In the nuclear ground state $T = 0$ and the total energy has a minimum (correspondingly $P = 0$) witht the following values of nucleon number density n_0

and binding energy per nucleon W_0:

$$n_0 = 0.16\,\mathrm{fm}^{-3}, \tag{82}$$

$$W_0 \equiv \frac{\epsilon(T=0, n=n_0)}{n_0} - m = -\,16\,\mathrm{MeV} \tag{83}$$

For the GCE, we have the transcendental equations:

$$P(T,\mu) = p^{id}(T,\mu^*) + n^2 u'(n), \tag{84}$$

$$\epsilon(T,\mu) = (1-bn)\epsilon^{id}(T,\mu^*) + nu(n) \tag{85}$$

$$n(T,\mu) = (1-bn)n^{id}(T,\mu^*) \tag{86}$$

$$\mu^* = \mu - bp^{id}(T,\mu^*) - u(n) - nu'(n) \tag{87}$$

where μ is the chemical potential, μ^* is the effective chemical potential, ϵ is the energy density, and the function $u(n)$ corresponds to the energy of attractive interactions per particle.

and for vdW specifically:

$$u(n) = -an, \tag{88}$$

therefore

$$u'(n) = -a. \tag{89}$$

Then:

$$P(T,\mu) = p^{id}(T,\mu^*) - an^2, \tag{90}$$

$$\epsilon(T,\mu) = (1-bn)\epsilon^{id}(T,\mu^*) - an^2. \tag{91}$$

From equation 86 at $T=0$ and $n=n_0$

$$n_0 = (1-bn_0)n^{id}(0,\mu^*) \tag{92}$$

So

$$n^{id}(0,\mu^*) = \frac{n_0}{1-bn_0} \tag{93}$$

On the other hand we have that the ideal fermi gas functions are defined as:

$$p^{\mathrm{id}}(T,\mu) = \frac{d}{6\pi^2}\int_0^\infty dk\,\frac{k^4}{\sqrt{m^2+k^2}}\left[\exp\left(\frac{\sqrt{m^2+k^2}-\mu}{T}\right)+1\right]^{-1},$$

$$n^{\mathrm{id}}(T,\mu) = \left(\frac{\partial p^{\mathrm{id}}}{\partial\mu}\right)_T,$$

$$\varepsilon^{\mathrm{id}}(T,\mu) = T\left(\frac{\partial p^{\mathrm{id}}}{\partial T}\right)_\mu + \mu n^{\mathrm{id}} - p^{\mathrm{id}}.$$

With the constraints we have that:

$$\text{At } T=0, \qquad f(k) = \left[\exp\left(\frac{\sqrt{k^2+m^2}-\mu}{T}\right)+1\right]^{-1} \longrightarrow \Theta(\mu - E_k), \qquad E_k = \sqrt{k^2+m^2}. \tag{94}$$

So the upper limit becomes the Fermi momentum

$$k_F = \sqrt{\mu^2-m^2}, \qquad \mu \geq m. \tag{95}$$

Then

$$p^{\mathrm{id}}(0,\mu) = \frac{d}{6\pi^2}\int_0^{k_F} dk\,\frac{k^4}{\sqrt{k^2+m^2}} = \frac{d}{48\pi^2}\left[\mu k_F(2\mu^2-5m^2) + 3m^4\ln\left(\frac{\mu+k_F}{m}\right)\right], \tag{96}$$

and

$$n^{\mathrm{id}}(0,\mu) = \frac{d}{6\pi^2}k_F^3. \tag{97}$$

Pluging eq. 92 into the later equation:

$$\frac{n_0}{1-bn_0} = \frac{d}{6\pi^2}k_F^3. \tag{98}$$

Solving for k_F

$$k_F(b) = \left(\frac{6\pi^2}{d}\frac{n_0}{1-bn_0}\right)^{1/3} \tag{99}$$

From eq 90 at $T=0$, and the constraint $P=0$

$$P(0,n_0) = p^{\mathrm{id}} - an_0^2 = 0 \quad \Rightarrow \quad a = \frac{p^{\mathrm{id}}}{n_0^2} \tag{100}$$

From eq **??** at $T=0$

$$\varepsilon(0,n_0) = (1-bn_0)\varepsilon^{\mathrm{id}} - an_0^2. \tag{101}$$

Divide by n_0:

$$\frac{\varepsilon(0,n_0)}{n_0} = (1-bn_0)\frac{\varepsilon^{\mathrm{id}}}{n_0} - an_0. \tag{102}$$

Applying the binding energy condition:

$$(1-bn_0)\frac{\varepsilon^{\mathrm{id}}}{n_0} - an_0 - m = -16. \tag{103}$$

Rearrange:

$$an_0 = (1-bn_0)\frac{\varepsilon^{\mathrm{id}}}{n_0} - (m-16), \tag{104}$$

$$a = \frac{1}{n_0}\left[(1-bn_0)\frac{\varepsilon^{\mathrm{id}}}{n_0} - (m-16)\right]. \tag{105}$$

From pressure 100 set equal:

$$\frac{p^{\mathrm{id}}}{n_0^2} = \frac{1}{n_0}\left[(1-bn_0)\frac{\varepsilon^{\mathrm{id}}}{n_0} - (m-16)\right]. \tag{106}$$

Multiply by n_0^2:

$$p^{\mathrm{id}} = (1-bn_0)\varepsilon^{\mathrm{id}} - n_0(m-16). \tag{107}$$

Then:

$$f(b) = p^{\mathrm{id}}(b) - (1-bn_0)\varepsilon^{\mathrm{id}}(b) + n_0(m-16) = 0. \tag{108}$$

Algorithm

1. Set the physical constants:
$$n_0 = 0.16\ \mathrm{fm}^{-3}, \qquad m \simeq 938\ \mathrm{MeV}, \qquad d = 4. \tag{109}$$
2. Choose a trial value of b.
3. Compute the effective ideal-gas density:
$$n_* = \frac{n_0}{1 - bn_0}. \tag{110}$$
4. Compute the corresponding Fermi momentum and Fermi energy:
$$k_F^* = \left(\frac{6\pi^2 n_*}{d}\right)^{1/3}, \qquad E_F^* = \sqrt{(k_F^*)^2 + m^2}. \tag{111}$$
5. Evaluate the ideal Fermi-gas functions:
$$\varepsilon^{\mathrm{id}}(b), \qquad p^{\mathrm{id}}(b). \tag{112}$$
6. Build the scalar function
$$f(b) = p^{\mathrm{id}}(b) - (1 - bn_0)\varepsilon^{\mathrm{id}}(b) + n_0(m - 16). \tag{113}$$
7. Solve numerically
$$f(b) = 0 \tag{114}$$
8. Once b is known, compute
$$a = \frac{p^{\mathrm{id}}(b)}{n_0^2}. \tag{115}$$
9. Construct the full zero-temperature pressure as a function of density:
$$P(0, n) = p^{\mathrm{id}}\left(\frac{n}{1 - bn}\right) - an^2. \tag{116}$$
10. Compute the derivative of the pressure at saturation:
$$\left(\frac{\partial P}{\partial n}\right)_{T=0, n=n_0}. \tag{117}$$
In practice, this is usually done numerically by a central difference:
$$\left(\frac{\partial P}{\partial n}\right)_{n_0} \approx \frac{P(0, n_0 + h) - P(0, n_0 - h)}{2h}. \tag{118}$$
11. Finally evaluate the incompressibility:
$$K_0 = 9\left(\frac{\partial P}{\partial n}\right)_{T=0, n=n_0}. \tag{119}$$

$$\boxed{\begin{array}{l} \text{Input: } n_0,\ m,\ d \\ \text{Solve } f(b) = 0, \\ f(b) = p^{\mathrm{id}}(b) - (1 - bn_0)\varepsilon^{\mathrm{id}}(b) + n_0(m - 16), \\ a = \dfrac{p^{\mathrm{id}}(b)}{n_0^2}, \\ K_0 = 9\left(\dfrac{\partial P}{\partial n}\right)_{T=0, n_0}. \end{array}} \tag{120}$$

From equation for 86 at ground state ($T = 0, P = 0, n = n_0 = -0.16\,\mathrm{fm}^{-3}, W_0 = -16\,\mathrm{MeV}$)

$$n(0,\mu) = (1 - bn(0,\mu))n^{id}(0,\mu^*) = n_0 = (1 - bn_0), n^{id}(0,\mu^*) \tag{121}$$

solving for n^{id}:

$$\frac{n_0}{(1 - bn_0)} = n^{id}, \tag{122}$$

now we can fix b and get n^{id}. On the other hand for pressure 84, we have:

$$P(0,\mu) = p^{id} + n^2 u'(n) = 0, \tag{123}$$

therefore solving for p^{id}

$$-n_0^2 u'(n) = p^{id}, \tag{124}$$

and $u'(n) \equiv u'(n, a, b)$, so with fix b we can numerically get a, but we still need to know p^{id} which from eq 5.1 we have at $T = 0$,

$$p^{id}(0,\mu) = \frac{d}{6\pi^2}\int_0^{kf} \frac{k^2}{\sqrt{k^2 + m^2}} \tag{125}$$

this can be computed computationally knowing that at ground state, for nuclear symmetric matter:

$$k_f = \left(\frac{6\pi^2 n^{id}}{d}\right)^{1/3} \tag{126}$$

with $d = 4$ and eq 122 we have:

$$k_f = \left(\frac{6\pi^2}{4}\frac{n_0}{(1 - bn_0)}\right)^{1/3} \tag{127}$$

So after integrating we can plug into the result.

To determine the parameter b, we impose the ground-state condition for symmetric nuclear matter at $T = 0$,

$$\frac{\varepsilon}{n_0} - m = -16 \text{ MeV}, \tag{128}$$

with

$$\varepsilon(0,\mu) = (1 - bn_0)\,\varepsilon^{\rm id}(0,\mu^*) + n_0\,u(n_0). \tag{129}$$

Using also

$$n_0 = (1 - bn_0)\,n^{\rm id}(0,\mu^*), \tag{130}$$

we divide by n_0 and obtain

$$\frac{\varepsilon}{n_0} = \frac{(1 - bn_0)\,\varepsilon^{\rm id}(0,\mu^*)}{n_0} + u(n_0) = \frac{\varepsilon^{\rm id}(0,\mu^*)}{n^{\rm id}(0,\mu^*)} + u(n_0). \tag{131}$$

Therefore, the binding-energy condition becomes

$$\frac{\varepsilon^{\rm id}(0,\mu^*)}{n^{\rm id}(0,\mu^*)} + u(n_0) - m = -16 \text{ MeV}. \tag{132}$$

Note that $\varepsilon^{\rm id}$ is the ideal Fermi-gas energy density, not the chemical potential. At $T = 0$, it is obtained from the Fermi integral

$$\varepsilon^{\rm id}(n^*) = \frac{d}{2\pi^2}\int_0^{k_F} k^2\sqrt{k^2 + m^2}\,dk, \tag{133}$$

where k_F is

$$k_F = \left(\frac{6\pi^2}{d}\,\frac{n_0}{1 - bn_0}\right)^{1/3}. \tag{134}$$

Thus, for each trial value of b, one computes n^{id}, then k_F, and evaluates $\varepsilon^{\rm id}(n^{id})$. The correct value of b is the one that satisfies

$$\frac{\varepsilon^{\rm id}(n^*)}{n^*} + u(n_0) - m = -16 \text{ MeV}, \tag{135}$$

together with the zero-pressure condition

$$p^{\rm id}(n^*) + n_0^2\,u'(n_0) = 0. \tag{136}$$